

Alden Cutler

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EDUCATION

Worcester Polytechnic Institute, Worcester, MA, 3.8 GPA Expected Graduation May 2026

B.S./M.S. in Computer Science, Specialization in Computer Security, Minor in Data Science

Awards: Dean's List (Fall 2023 – Present), Presidential Scholarship (4-year, \$21,000 per year), Winner – GoatHacks 2024

Seattle Academy of Arts and Sciences, Seattle, WA, 3.96 wGPA Graduated June 2022

Awards: National 2nd Place – AFA Cyberpatriot Competition Gold Division, 1st Place - FRC Rookie of the Year 2019-2020, 1st Place – Think Award at FTC World Championships 2018-2019, 2nd Place – Oregon State Fair FRC Event 2022

WORK EXPERIENCE

Tesla, *Software Engineer Intern*, Palo Alto, CA May 2025 – August 2025

- Reduced CI/CD infrastructure costs by 2.5x through component-specific firmware builds.
- Saved 30 hours of manual work per quarter by automating quantitative analysis of firmware releases using Jenkins, Python, and Splunk enterprise. New system gets results every week instead of once per quarter.
- Improved internal firmware release tooling which fixed a misreporting metric on over 400,000 Tesla Energy devices.
- Reduced time to create firmware release sign-off checklist by 80% with Jenkins automation and Python.
- Wrote firewall config for Megapack internal computer, removing a blanket accept on all customer I/O ports.

Boeing, *Cybersecurity Intern*, Seattle, WA March 2022 – May 2022

- Used shell scripting, Python, and statistical analysis to develop software that monitors per-process resource usage and establishes standard baselines for the aircraft's onboard computer.

LEADERSHIP EXPERIENCE

Worcester Polytechnic Institute & Microsoft, *Major Qualifying Project* August 2026 – March 2026

- Built a domain-specialized Microsoft Intune chatbot by optimizing a RAG pipeline with hybrid retrieval (semantic embeddings + BM25/RRF in PostgreSQL/PGVector).
- Engineered agent-based quality controls (context sufficiency and response review loops).
- Deployed as an OpenAI-compatible FastAPI service with citation-aware responses and streaming.
- Evaluated pipeline against baseline RAG variants using Azure AI Evaluation (Groundedness, Relevance, Retrieval).
- Co-authored a released capstone report on trustworthy, domain-specific question answering, including methodology, tradeoffs, and future improvements.

Worcester Polytechnic Institute & Brigham and Women's Hospital, *Lead Developer* March 2025 – May 2025

- Led a 10-person Agile team building a web application prototype for Brigham & Women's Hospital.
- Developed using PostgreSQL, Express, React, Node.js stack with Typescript, PrismaORM, MaterialUI, and Tailwind.
- Deployed to the cloud using AWS EC, Docker, and AWS RDS to run the PostgreSQL server.
- Implemented hospital pathfinding, map editor, integrated service request modules, and voice activated user interface.

Worcester Polytechnic Institute, *Interactive Qualifying Project*, Lyon, France May 2024 – July 2024

- Led research efforts to use data scraping, Python/Jupyter notebooks, and a SQL database of >500,000 data points to quantitatively analyze efficiency and spatial density of public transit in Lyon, France.
- Conducted stakeholder interviews and translated findings into actionable insights for global transit authorities.

SKILLS AND INTERESTS

- *Programming Languages*: Python, JavaScript, TypeScript, C, C++, R, Java, x86 Assembly
- *Software*: Git, GitHub, React.js, Node.js, Tailwind.css, PostgreSQL, Jenkins, Confluence, Ghidra, GDB, GEF, Unix/Linux OS, PrismaORM, MongoDB, Flask, Nmap, Pandas, Numpy, PyTorch
- *Language Skills*: English (Native), French (Fluent), Mandarin Chinese (Limited Working Proficiency)
- *Interests*: Cybersecurity, Software Development, Artificial Intelligence